

Quantifying the prevalence of energy poverty across Canada: Estimating domestic energy burden using an expenditures approach

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Key messages

- Research on energy poverty in Canada is an understudied area in Canadian scholarship.
- Energy poverty is studied by examining the ratio of household energy expenditures to household income.
- Households in energy poverty spent more on energy expenditures as a proportion of their total expenditures compared to households not in energy poverty.

Energy poverty is gaining public attention in Canada. Based on statistical analysis of Statistics Canada's 2016 Survey of Household Spending, we estimate that 7–9% of households spent more than 10% of their income on energy expenditures. Households in the Atlantic provinces faced the most energy poverty and highest energy expenditures. Low income, geography, and dwelling conditions were the main predictors of energy poverty. Households in energy poverty also spent approximately five times more on energy expenditures as a proportion of their total expenditures compared to households not in energy poverty. Our study fills a key research gap as a limited number of studies exist on the extent and drivers of energy poverty in Canada. Further, our study's robust quantitative estimates allow benchmarking and comparative research. These estimates lay the groundwork for evidence-based solutions—and our findings highlight the need to reconsider current policies. Considering the challenge of the ratio of energy costs to income is vital, especially in the aftermath of events such as COVID-19 or the 2021 heatwave in western Canada, which result in different energy behaviours and needs. More broadly, in the regular day-to-day, energy services should be seen as necessary and decision makers ought to consider the energy burden of Canadians.

Keywords: energy poverty, energy burden, household energy expenditure, logistic regression, Engel curve

La prévalence de la pauvreté énergétique au Canada : L'estimation du poids financier de l'énergie domestique selon l'approche axée sur les dépenses

Le phénomène de pauvreté énergétique attire de plus en plus l'attention au Canada. À partir des données de l'Enquête sur les dépenses des ménages de Statistique Canada (2016), nous estimons qu'entre 7 et 9% des ménages ont engagé plus de 10% de leurs revenus en dépenses énergétiques. Selon nos analyses, les ménages

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des provinces de l'Atlantique connaissent la plus importante pauvreté énergétique et les dépenses énergétiques les plus élevées. De plus, le faible revenu, la localisation et les conditions de logement sont les principales variables explicatives de la pauvreté énergétique. Finalement, les ménages en situation de pauvreté énergétique ont engagé environ cinq fois plus d'argent en dépenses énergétiques proportionnellement à leurs dépenses totales, si on les compare aux autres ménages. En conséquence, notre étude comble une lacune de la recherche puisqu'il n'existe qu'un nombre limité d'études sur la portée et les facteurs de la pauvreté énergétique au Canada. Par ailleurs, les traitements quantitatifs approfondis de notre enquête permettent de faire un étalonnage et une démarche comparative. Ces résultats jettent les bases de solutions fondées sur des données probantes car nos conclusions mettent également en évidence la nécessité de réexaminer les politiques actuelles. Au final, il est essentiel de reconnaître l'importance du problème du ratio des coûts énergétiques, particulièrement à la suite d'événements tels que la COVID-19 ou les canicules, lesquels ont eu pour effet de modifier les comportements et les besoins énergétiques.

Mots clés : pauvreté énergétique, poids financier de l'énergie, dépenses énergétiques des ménages, régression logistique, courbe d'Engel

Introduction

Energy poverty continues to grow in recognition and prevalence (Belaïd 2018); millions of people live in energy poverty, including in industrialized nations (Bouzarovski et al. 2012; Day et al. 2016). Energy poverty definitions are debated but there is consensus that, regardless of geography or politics, “energy poverty” refers to “the inability to attain a socially and materially necessitated level of domestic energy services” (Bouzarovski and Petrova 2015, 31). Energy services like heating, cooling, lighting, cooking, transportation, and access and connectivity to information are essential (González-Eguino 2015). Going without can impact mental health, increase social stress, and reduce emotional resiliency (Liddell and Morris 2010; Dear and McMichael 2011; O'Sullivan et al. 2017). Energy poverty has also been linked to deaths, asthma, respiratory illnesses, arthritis, rheumatism, and thermal discomfort (Wright 2004; Rudge and Gilchrist 2005; Hernández and Bird 2010; Heyman et al. 2011; Ormandy and Ezratty 2012; Chen et al. 2017). Additionally, research has found that increases in household heating prices have previously contributed to food insecurity in Canada and thus to the phenomenon of “heat or eat” (Emery et al. 2012). Further, energy poverty has been shown to vary according to geography (Bouzarovski and Simcock 2017).

Energy poverty in Canada has been gaining recognition (albeit slowly), but the issue has been overlooked in Canadian research, unlike in other countries such as Austria, Australia, France, Germany, Greece, Hungary, Italy, New Zealand, Ukraine, the United Kingdom (UK), and the United States (US) (e.g., Chester and Morris 2011; Boardman 2012;

Brunner et al. 2012; Dubois 2012; Herrero and Ürge-Vorsatz 2012; Petrova et al. 2013; Miniaci 2014; Tews 2014; Lawson et al. 2015; Teller-Elsberg et al. 2016; Ntaintasis et al. 2019). The limited research related to energy poverty in the Canadian context has focused on: the built environment (Tardy and Lee 2019); energy planning and access in two Indigenous communities (Rezaei and Dowlatabadi 2015; Rezaei 2017); electricity infrastructure and demand in remote communities (Government of Canada 2011); energy affordability and energy efficiency for low-income households (Energy Efficiency Working Group 2008); energy transitions (Rowlands 2015; Rowlands and Gord 2016); and research and knowledge mobilization in less academic venues (McEachern and Vivian 2010; Lee et al. 2011; Green et al. 2016; CUSP 2019).

The purpose of this paper is fourfold. First, it fills a research gap; limited academic study exists on the extent and drivers of energy poverty in Canada, which this paper begins to empirically address using Statistics Canada's *Survey of Household Spending* (SHS). Second, its robust quantitative estimates of energy poverty allow benchmarking and comparative research. Third, these estimates lay the groundwork for evidence-based solutions and, fourth, they have implications for other policy areas.

Background and context

Energy poverty overview

Energy poverty is now an officially recognized form of deprivation in many countries (European Commission, n.d.), though several terms are used

interchangeably (Thomson et al. 2016). Historically, “energy poverty” was used in reference to energy access in less developed or industrialized countries (Nussbaumer et al. 2012), whereas “fuel poverty” was used in relation to energy affordability in the UK (Boardman 1991). Increasing energy market globalization, as well as research on energy service access in the more industrialized countries, challenge the energy-fuel poverty binary (Bouzarovski and Petrova 2015). We use the term “energy poverty” herein (unless referring to specific research or reports) to align our language with the dominant usage in Canada (e.g., Environmental Commissioner of Ontario 2018).

Richardson (1978) was among the first to characterize energy poverty as a situation wherein people lack resources to afford costs associated with heating, lighting, and cooking. Lewis (1982) also defined energy poverty as households’ difficulties affording adequate heat. Early study of energy poverty in the UK started in the 1980s with Bradshaw and Hutton (1983). In 1991, energy poverty was given a more formal definition based on “energy burden” or percentage of household income spent on home energy services (Bednar and Reames 2020; Brown et al. 2020): households needing to spend more than 10% of their income maintaining comfortable indoor temperatures are considered to be in energy poverty (Boardman 1991). This definition has been criticized for insufficiently capturing the scale and nature of energy poverty (Moore 2012), but it nevertheless provides a starting point for energy poverty investigations, and energy burden is a “dominant construct” used by researchers (Brown et al. 2020, 1). It is relevant to note that “energy poverty is distinct from wider forms of poverty due to the role of domestic and networked energy infrastructures in its manifestation” (Robinson et al. 2019, 1190)—in other words, energy poverty is not only linked to monetary issues of high energy prices and low incomes, but also to infrastructure issues such as the energy efficiency of buildings. In the next sections we situate energy poverty within geography-based frameworks followed by an overview of household energy use in Canada, including background on household energy costs, income, and housing specific to Canada.

Energy poverty as a spatial phenomenon

Indeed energy poverty is “deeply” driven by energy prices and household income—given their combination is associated with the affordability of energy—however drivers can vary at multiple scales (Bouzarovski and Simcock 2017). For example, household income and energy prices vary by regions and nations. Geographic segregation also exists at the neighbourhood level, with income poverty more prevalent in certain neighbourhoods than in others (Bouzarovski and Simcock 2017). Neighbourhood patterns of low-income can partly be associated with the clustering of apartment housing, visible minority groups, and levels of racial diversity (e.g., Walks and Bourne 2006).

The manner in which energy is accessed can be tied to the “geographical assemblage” of energy flows, production and distribution infrastructures, built environment characteristics, and social and economic networks (Harrison and Popke 2011). At the household level, characteristics such as heating systems, the energy carriers used with them, and the energy efficiency of a dwelling, also create particular ways of accessing and affording energy services (Maxim et al. 2017). Environmental features such as climate conditions underpinned by spatial and temporal differences additionally interact with the geographical assemblage (Bouzarovski and Simcock 2017). Together, dynamic features varying across space and time are important when considering the relationship people have with their energy services given that differences can manifest due to spatial inequities, and put some in more precarious and vulnerable positions than others (Yenneti et al. 2016).

“Landscapes” of household energy use, energy costs, income, and housing

Canadian households use energy for many purposes: 2% on space cooling, 3% on lighting, 13% on appliances, 18% on water heating, and 64% on space heating (Natural Resources Canada 2021). While electricity is the energy source for many household activities, natural gas is dominant for space and water heating across the country (Natural Resources Canada, n.d.). These forms of energy have different properties, with natural gas generally being more efficient and cost effective than electricity for heating (Harrison and Popke 2011).

Given Canada's geography and climate, sufficient and comfortable indoor temperatures are very important. A snapshot of temperatures in Toronto provides an example of extremes and variability: in 2018, January temperatures averaged -10°C with the coldest day falling to -24°C, while July temperatures averaged 29°C with the hottest day reaching 34°C (Government of Canada n.d.b., n.d.c.). According to Tardy and Lee (2019, 47), "beyond mere comfort, home heating in Canada is literally a matter of life and death." The 2021 summer heat wave in western Canada also points to cooling becoming a matter of life and death.

In Canada, the divergence between energy costs and household income appears to be increasing. Across Canada, electricity prices (as well as energy sources for generation) vary by province and an average monthly electricity bill, based on a monthly consumption of 1000 kWh, can range from \$73 in Quebec to \$382 in the Northwest Territories; the 2021 Canadian average was \$179 (Statista 2021). As shown in Figure 1, household energy demand is higher for higher-income groups; as household income increased in 2015, energy expenditures on electricity and natural gas tended to also increase. The two lowest income groups (under \$20,000, and \$20,000 to \$39,000) used approximately 9000 kWh of electricity and 80 GJ of natural gas, while the

highest income group (\$100,000 and over) consumed around 14,000 kWh and over 90 GJ.

Housing costs also impact disposable income for energy expenditures. Across Canada, many households struggle with affordable and adequate housing (e.g., Ley et al. 2020). Housing has reached a critical point; the federal government recently presented its \$40 billion National Housing Strategy (Government of Canada n.d.a.). The Canada Mortgage and Housing Corporation (2018) considers housing to be affordable when it costs less than one-third of a household's before-tax income, while the BC Non-Profit Housing Association (BCNPHA) uses rent and energy expenditures (i.e., utility costs) together as an affordability indicator (BCNPHA, n.d.). According to the BCNPHA's index, 40% of Canadians are spending over 30% of their income on rent and utilities, and those in each province's lowest income group spend a greater share of income on rent and utilities, for example: 80% in British Columbia (\$0 to \$23,378), 60% in Alberta (\$0 to \$33,002), and 73% in Ontario (\$0 to \$22,385) (BCNPHA, n.d.). Spending on rent and utilities also differs between single mothers, single fathers, couples without children, and couples with children, with single mothers experiencing the largest burden (BCNPHA, n.d.).

With respect to dwellings, energy performance varies by structural characteristics. A positive

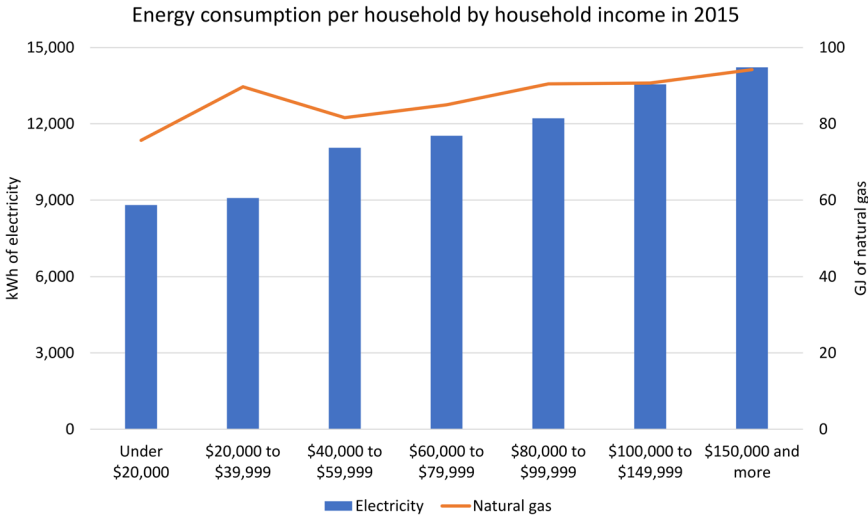


Figure 1
Energy consumption per household by household income in 2015.
SOURCE: Statistics Canada (2015).

relationship between dwelling age and household energy use has been found in every province, showing older dwellings are less energy efficient (Tardy and Lee 2019). Differences also exist across housing type. For example, community housing is associated with higher levels of energy use due to the sector's aging buildings (75% built prior to 1990) and widespread deferred maintenance (Government of Canada n.d.a.; Statistics Canada 2018). In British Columbia, per unit energy consumption in community housing is 60% higher than in market housing (BCNPHA 2019).

Methodology

Different quantitative and qualitative approaches are used to measure and define energy poverty, with data availability often guiding choice of methodology (Brown et al. 2020). We used the expenditure approach, taking into account households' energy costs against a threshold to estimate energy burden, which is then used as a proxy for energy poverty (Brown et al. 2020). While we recognize the need to consider various approaches (Hills 2012; Heindl 2015), the present study focuses on an income-based measure due to limited data availability. We also used Engel curves to study the relationship between household expenditures on energy services and household income. "Engel curves describe how household expenditure on particular goods or services depends on household income" (Chai and Moneta 2010, 225), and have been used in studies of issues concerning Canadian public policy (e.g., Emery and Guo 2020). In his seminal study on the costs of living among working families, Engel (1895) argued that the income of a population must be, at minimum, high enough to cover needs and costs of living. Overall, our methodological approach is useful for energy and income-related policy.

Data

We drew on confidential microdata collected from January to December 2016 for Statistics Canada's national 2016 SHS; these were the most recent data available at the time of analysis. This survey is conducted annually in Canada's provinces and every other year in the territories. Survey questions relate to household expenditures, dwelling characteristics, household demographics, among others. Income

and income tax data come from personal tax return files from the Canada Revenue Agency. The SHS conducts interviews to administer a questionnaire with varying recall periods based on type of expenditure. This is combined with an expenditure diary that certain households ($n = 3809$) complete for two weeks post-interview.

The SHS is a sample survey with a cross-sectional design. A stratified multi-stage sampling design was used to select the sample in stages. The first stage is a sample of geographic areas (clusters) and the second consists of sampling dwellings within each cluster. Selected dwellings constitute the sample of households within the target population—the population of Canada's provinces, excluding residents of institutions, members of the Canadian Forces living in military camps, and people living on "Indian reserves"; exclusions form approximately 2% of the population. The 2016 SHS original sample consisted of 17,590 households; a response rate of 65.1% resulted in 11,446 households in the final sample. Data are weighted to compensate for the probability of selection and to match Canadian households to census data; statistical estimates are presented on a population scale.

Measurement and analysis

We first constructed a measure of energy poverty using the 10% approach (Hills 2012). According to this indicator, a household for which the share of energy expenditures as a proportion of household income is greater than 10% is said to be in energy poverty (Teller-Elsberg et al. 2016). We used actual energy expenditures (versus *required* expenditures to maintain a certain temperature) given the non-availability of required expenditures data. We examined household spending on electricity, natural gas, other fuel, heating oil, propane, and wood. We also adjusted for housing costs by subtracting mortgage or rent payments from household income. Experts argue taking housing costs into account provides a clearer picture of disposable income and the affordability of bills, since housing costs are typically prioritized compared to other costs (Hills 2012; Moore 2012). Housing costs are also geographically variable so estimating income before housing costs can produce misleading estimates of the extent and jurisdictional distribution of energy poverty (Legendre and Ricci 2015). It should be noted that household income in our dataset is

measured before tax; after tax data were unavailable. Therefore our two indicators are:

- 1) Energy poverty = $[(\text{electricity} + \text{natural gas} + \text{other fuel} + \text{heating oil} + \text{propane for heating and cooking} + \text{wood and other fuel for heating and cooking}) / (\text{annual household income})] > 0.10$
- 2) Energy poverty adj. for housing costs = $\{[(\text{electricity} + \text{natural gas} + \text{other fuel} + \text{heating oil} + \text{propane for heating and cooking} + \text{wood and other fuel for heating and cooking}) / ((\text{annual household income}) - (\text{rent or mortgages}))] > 0.10$

Our analysis consists of univariate and bivariate analyses to describe the dependent and independent variables. As our dependent variables are dichotomous measures (i.e., experiencing energy poverty or not experiencing energy poverty), two binary logistic regressions were conducted to examine the factors that influence the likelihood of experiencing energy poverty in Canada. Methods align with previous quantitative energy poverty study (e.g., Thomson and Snell 2013; Belaïd 2018). Independent variables are based on existing literature and quantitative explanations of energy poverty focused on costs borne by households and dwelling characteristics. We also considered various demographics and examined a broad range of factors to identify which ones may increase the likelihood of energy poverty. A list of variables and their units of measurement are shown in Table 1.

For developing the Engel curves we first removed all observations with negative and zero values of energy expenditures and annual household income. Second, to mask individual observations and protect confidentiality, we aggregated individual-level observations by taking the average of every five observations to generate scatter points. We then plotted a scatterplot with the energy budget share (i.e., the proportion spent on energy expenditures out of total expenditures) on the vertical axis and household income on the horizontal axis.

Variance inflation factors and tolerance levels were assessed with respect to suggested values in the literature given their potential to bias parameter estimates; results were within limits. Missing data were carefully examined and were deemed not a problem. We excluded a handful of cases having household income less than \$217,462, equal to 0, or greater than \$200,000 from the analysis, as they were deemed outliers. Applied survey weights were

standardized to prevent estimates from being inflated. Analyses were carried out in SPSS and R.

Study limitations

Our study has some limitations. First, Canadian data do not exist that account (together) for variables such as energy expenditures, energy consumption, temperature set points, various demographic information, etc. Second, data are from 2016; newer data might yield different results. Third, the 10% indicator may not capture the extent of energy poverty, that is, number of households in energy poverty; households spending less than 10% due to financial strain and budgeting may in fact experience energy poverty (Hills 2012). Nor does this indicator capture the depth of the problem (i.e., severity of experience), which could be measured with qualitative indicators (Heindl 2015). The 10% indicator is also sensitive to energy pricing; energy poverty can be underestimated when prices are low and overestimated when prices are high (Romero et al. 2018). We emphasize our study is a starting point; we welcome further research on energy poverty in Canada using different data and indicators.

Results

Results show that approximately 7% of or 900,863 Canadian households spent more than 10% of their income on energy expenditures in 2016. As shown in Figure 2, energy poverty varied across the country, ranging from 4% in British Columbia to an average of roughly 19% in the Atlantic provinces (New Brunswick, Nova Scotia, Prince Edward Island, Newfoundland and Labrador). Adjusting for housing costs, energy poverty in each province increased, bringing national energy poverty to 9%, $n = 1,215,263$.

Median annual household income in Canada for those not in energy poverty was estimated at \$68,989; for those in energy poverty it was \$22,407. Levels of income varied between provinces, but lower incomes did not dominate any particular geography. The major source of income for all Canadians was employment (62%), while for households in energy poverty it was government transfer payments (68%). Among all households, 75% of household members were full-time earners compared to 24% in an energy poverty household. Households in energy poverty spent nearly

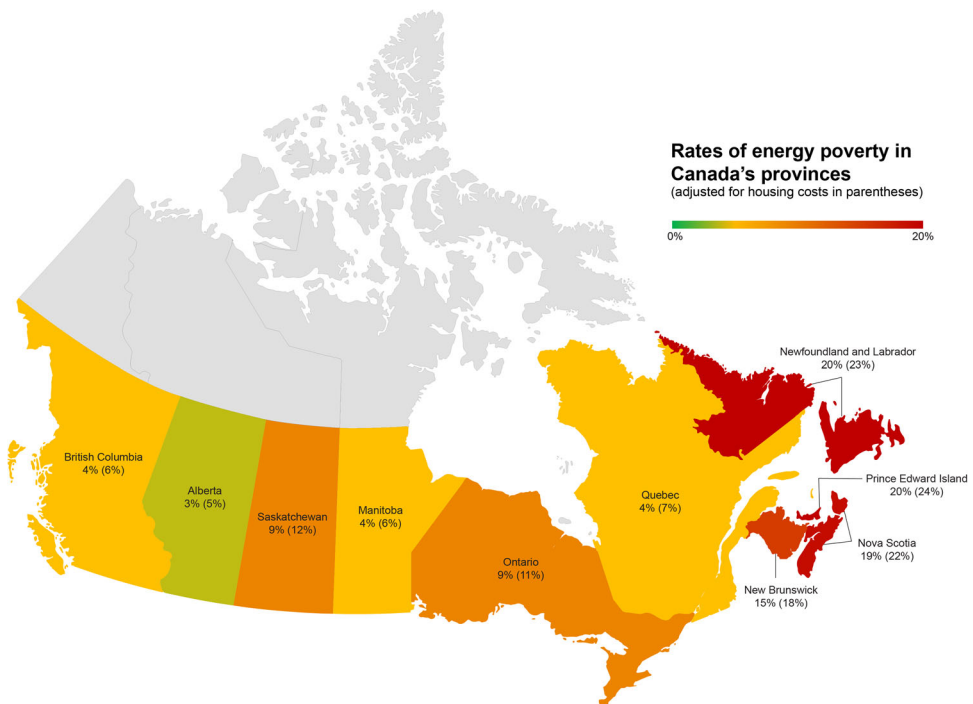
Table 1
List of variables and measurement

Variable	Measurement
Province of residence	1 = Ontario; 2 = Alberta; 3 = British Columbia; 4 = Manitoba; 5 = New Brunswick; 6 = Newfoundland and Labrador; 7 = Nova Scotia; 8 = Prince Edward Island; 9 = Quebec; 10 = Saskatchewan
Lives in a rural area	1 = Yes; 0 = No
Tenure type	1 = Own with mortgage; 2 = Own without mortgage; 3 = Rented
Sex	1 = Male; 0 = Female
Age	Measured in years
Marital status	1 = Married; 2 = Living common-law; 3 = Divorced; 4 = Separated; 5 = Single, never married; 6 = Widowed
Highest level of educational attainment	1 = High school or below; 2 = Some post-secondary; 3 = Bachelor's degree or above
Household type	1 = One-person household; 2 = Parents with children; 3 = Other
Size of household	Number of household members
Number of children	Number of children aged 14 or under
Internet access	1 = Yes; 0 = No
Health care costs (i.e., direct costs to household and private health insurance plan premiums)	Measured in dollars
Major health care equipment costs (i.e., hearing aids, wheelchairs and scooters, hospital beds, patient lifts, and CPAP machines)	Measured in dollars
Major source of income	1 = Employment; 2 = Government transfer payments; 3 = Other
Employment insurance	1 = Yes; 0 = No
Full-time earners	Number of full-time earners in the household
Part-time earners	Number of part-time earners in the household
Full-time work weeks	Number of weeks worked full-time with pay
Part-time work weeks	Number of weeks worked part-time with pay
Private use vehicle	1 = Yes; 0 = No
Private use vehicle cost	Measured in dollars
Public transit cost	Measured in dollars
Dwelling type	1 = Single detached; 2 = Double (semi-detached); row or terrace (townhouse); duplex; 3 = Low-rise and high-rise apartments
Construction period	1 = 1960 or before; 2 = 1961–1980; 3 = 1981–2010; 4 = 2011 or after
Dwelling needs repair	1 = No; 2 = Minor repair; 3 = Major repair
Repairs and maintenance – Interior walls and ceilings	Measured in dollars
Repairs and maintenance – Caulking and weather stripping	Measured in dollars
Repairs and maintenance – Insulation	Measured in dollars
Repairs and maintenance – Roofs	Measured in dollars
Repairs and maintenance – Exterior walls, siding, soffits, facia, chimneys and foundations	Measured in dollars
Repairs and maintenance – Windows and exterior doors	Measured in dollars
Repairs and maintenance – Heating and/or air conditioning equipment	Measured in dollars
Tenants' repairs and improvements	Measured in dollars
Total expenditure for refrigerators and freezers	Measured in dollars
Total expenditure for microwave ovens	Measured in dollars
Total expenditure for cooking appliances	Measured in dollars
Total expenditure for washers and dryers	Measured in dollars
Total expenditure for dishwashers	Measured in dollars
Total expenditure for room air conditioners, portable humidifier and dehumidifier	Measured in dollars

SOURCE: 2016 *Survey of Household Spending*.

double (\$154) the average (\$67) on major health care equipment, and spent less on private vehicle (\$2447) and public transit costs (\$110) than those not in energy poverty (\$4580 and \$271,

respectively). Annual median spending on energy expenditures for non-energy poverty households was estimated at \$1800; for energy poverty households it was \$3623. Notably, on average, energy

**Figure 2**

Rates of energy poverty in Canada's provinces.

SOURCE: Author's own calculation from the 2016 *Survey of Household Spending*.

expenditures were highest in the Atlantic provinces.

Of those considered to be in energy poverty, 24% reported living in a rural area compared to 10% of all Canadians living in rural areas. Among all households, 64% reported owning their home and 36% were renting, in contrast with 71% owning and 27% renting among those in energy poverty. Of households in energy poverty, 66% reported living in homes constructed prior to 1980 (compared to 55% of those not in energy poverty), and 39% reported that their dwellings needed minor or major repair (average for not in energy poverty was 30%).

Interestingly, results show substantial differences in the energy poverty rate by sex: more female (62%) than male (38%) were in energy poverty. Further, more energy poverty households were one-person households (53%) than not energy poverty households (29%); living alone has been associated with financial difficulties, increasing the likelihood of being in energy poverty (Legendre and Ricci 2015). Levels of education also varied—in energy-poor households

50% reported having high school education or below, 31% some post-secondary, 18% a bachelor's degree or above, compared to 36%, 36%, and 28%, respectively for those not in energy poverty. All descriptive statistics can be found in Table 2; figures in text may not match exactly due to rounding.

Next we present exponentiated coefficients (i.e., odds ratios) from the logistic regression models that are statistically significant. We present unique effects and therefore all other model variables are controlled for during discussion of a particular variable (even if not explicitly stated). Odds ratios shown in Table 3 reflect the odds a household with a particular characteristic will be in energy poverty relative to a household in the reference category. An odds ratio of 1 indicates households with that characteristic are equally likely to experience energy poverty as those in the reference category. An odds ratio greater than 1 indicates a higher risk of energy poverty, while a ratio below 1 indicates lower risk.

Overall, the independent and control variables explain approximately 17% and 11% of Canada's

Table 2
Sample descriptive statistics

Variable	Categories	Full sample		Energy poverty		Energy poverty adjusting for housing costs	
		N = 13,259,915 % or Mean (S.D.)		No N = 12,359,051 % or Mean (S.D.)	Yes N = 900,864 % or Mean (S.D.)	No N = 12,045,083 % or Mean (S.D.)	Yes N = 1,214,832 % or Mean (S.D.)
Province of residence	Ontario	37.1%		36.4%	46.3%	36.4%	43.6%
	Alberta	10.0%		10.4%	4.7%	10.5%	5.6%
	British Columbia	13.3%		13.7%	8.1%	13.8%	8.8%
	Manitoba	3.5%		3.6%	2.1%	3.6%	2.1%
	New Brunswick	2.3%		2.1%	5.2%	2.1%	4.6%
Lives in a rural area (less than 1000)	Newfoundland and Labrador	1.5%		1.3%	4.6%	1.3%	3.8%
	Nova Scotia	2.9%		2.5%	8.0%	2.5%	7.0%
	Prince Edward Island	0.4%		0.4%	1.3%	0.4%	1.2%
	Quebec	25.9		26.6%	15.4%	26.5%	19.4%
	Saskatchewan	3.0%		2.9%	4.2%	2.9%	4.0%
Tenure type	Yes	10.4%		9.4%	24.1%	9.4%	20.5%
	No	89.6%		90.6%	75.9%	90.6%	79.5%
	Own with mortgage	35.9%		36.2%	26.6%	36.0%	31.8%
	Own without mortgage	28.6%		27.2%	44.1%	27.9%	32.7%
	Rented	35.5%		35.7%	27.3%	35.3%	33.9%
Sex	male	47.8%		48.5%	38.2%	48.7%	38.7%
	Female	52.2%		51.5%	61.8%	51.3%	61.3%
Age	52.01(17.26)			51.66 (17.17)	56.89 (17.77)	51.73 (17.22)	54.79(17.47)
	Married	43.6%		44.8%	27.7%	44.9%	30.9%
	Living common-law	12.6%		13.0%	7.0%	13.1%	7.8%
	Divorced	9.3%		9.0%	12.5%	8.9%	12.9%
	Separated	5.3%		5.0%	10.2%	5.0%	9.0%
Highest level of educational attainment	Single, never married	20.2%		19.9%	23.4%	19.8%	24.0%
	Widowed	9.0%		8.2%	19.1%	8.3%	15.4%
	High school or below	37.0%		35.9%	50.4%	35.3%	51.8%
	Some post-secondary	36.4%		36.4%	31.3%	36.5%	31.2%
	Bachelor's degree or above	26.9%		27.5%	17.9%	27.8%	16.7%
Household type	One-person household	30.4%		28.7%	52.5%	28.8%	46.2%
	Parents with children	30.4%		31.1%	20.8%	40.3%	28.9%
	Other	39.2%		40.2%	26.7%	31.0%	25.0%
	Size of household	2.39 (1.40)		2.43 (1.40)	1.84(1.24)	2.42 (1.40)	2.02(1.35)
	Number of children	0.38 (0.83)		0.39 (0.84)	0.26 (0.73)	0.39 (0.83)	0.34(0.81)
Internet access	Yes	86.6%		87.5%	74.5%	87.7%	75.1%
	No	13.4%		12.5%	25.5%	12.3%	24.9%
Health care costs (i.e., direct costs to household and private health insurance plan premiums)	2048.60 (2763.46)			2705.04(2692.66)	1685.87 (3578.70)	2087.34 (2792.54)	1664.51(2424.32)

Table 2
(Continued)

Variable	Categories	Full sample		Energy poverty		Energy poverty adjusting for housing costs	
		N = 13,259,915 % or Mean (S.D.)	No N = 12,359,051 % or Mean (S.D.)	Yes N = 900,864 % or Mean (S.D.)	No N = 12,045,083 % or Mean (S.D.)	Yes N = 1,214,832 % or Mean (S.D.)	
Major health care equipment costs (i.e., hearing aids, wheelchairs and scooters, hospital beds, patient lifts, and CPAP machines)		67.30(471.62)	60.98 (444.53)	153.96 (745.21)	62.65 (451.16)	113.43(638.43)	
Major source of income	Employment	62.3%	64.8%	28.4%	64.9%	36.4%	
	Government transfer payments	34.5%	32.0%	67.8%	32.0%	58.8%	
	Other	3.2%	3.2%	3.8%	3.1%	4.8%	
Employment insurance	Yes	16.3%	16.9%	8.9%	16.8%	12.3%	
	No	83.7%	83.1%	91.1%	83.2%	87.8%	
Full-time earners		0.75(0.80)	0.79(0.81)	0.24(0.49)	0.79(0.81)	0.35(0.62)	
Part-time earners		0.53(0.75)	0.55(0.76)	0.39(0.62)	0.54(0.76)	0.45(0.68)	
Full-time work weeks		24.39(24.52)	25.53(24.52)	8.73(18.35)	25.56(24.54)	12.73(21.07)	
Part-time work weeks		5.12(13.87)	4.99(13.70)	6.93(15.99)	5.02(13.74)	6.14(15.14)	
Private use vehicle	Yes	82.7%	83.1%	77.1%	83.6%	74.1%	
	No	17.3%	16.9%	22.9%	16.4%	25.9%	
Private use vehicle cost		4435.23(11809.05)	4580.16(12011.50)	2446.85(8314.50)	4645.93(12118.64)	2346.08(7826.89)	
Public transit cost		259.91(627.74)	270.82(640.62)	110.18(382.12)	272.65(644.24)	133.56(410.47)	
Dwelling type	Single-detached	52.6%	50.4%	68.8%	50.2%	65.7%	
	Double (semi-detached); row or terrace	17.3%	30.8%	14.2%	31.0%	16.0%	
	(townhouse); duplex						
Construction period	Low-rise and high-rise apartments	30.2%	17.2%	13.5%	17.1%	15.8%	
	1960 or before	24.7%	30.9%	37.0%	30.8%	36.8%	
	1961–1980	30.8%	23.8%	28.6%	23.5%	30.1%	
	1981–2010	38.9%	39.6%	29.5%	39.8%	30.3%	
	2011 or after	5.6%	5.7%	4.8%	5.9%	2.8%	
Dwelling needs repair	No	69.6%	70.3%	61.3%	70.8%	58.1%	
	Minor repair	8.2%	7.8%	13.1%	7.7%	12.9%	
	Major repair	22.2%	21.9%	25.6%	21.5%	29.0%	
Repairs and maintenance–Interior walls and ceilings		37.10(468.52)	38.55(477.27)	17.09(324.91)	39.03(482.80)	17.96(290.68)	
Repairs and maintenance–Caulking and weather stripping		5.66(65.42)	5.69(67.13)	5.31(34.13)	5.59(66.10)	6.44(58.21)	
Repairs and maintenance–Insulation		3.59(39.01)	3.20(35.99)	8.96(67.90)	3.20(36.24)	7.42(59.83)	
Repairs and maintenance–Roofs		19.02(236.64)	18.68(237.61)	23.72(222.92)	18.87(239.86)	20.53(201.99)	
						(Continued)	

(Continued)

Table 2
(Continued)

Variable	Categories	Full sample N = 13,259,915 % or Mean (S.D.)	Energy poverty		Energy poverty adjusting for housing costs	
			No N = 12,359,051 % or Mean (S.D.)	Yes N = 900,864 % or Mean (S.D.)	No N = 12,045,083 % or Mean (S.D.)	Yes N = 1,214,832 % or Mean (S.D.)
Repairs and maintenance—Exterior walls, siding, soffits, fascia, chimneys and foundations		46.69(648.03)	42.85(606.19)	99.47(1066.68)	43.60(613.70)	77.41(921.42)
Repairs and maintenance—Windows and exterior doors		8.02(179.04)	7.75(168.84)	11.70(284.28)	7.93(170.99)	8.93(245.06)
Repairs and maintenance—Heating and/or air conditioning equipment		21.04(192.85)	21.19(196.14)	19.05(140.05)	21.62(198.60)	15.36(121.89)
Tenants' repairs and improvements		42.33(379.20)	38.78(360.86)	91.12(570.50)	38.41(355.14)	81.24(564.50)
Total expenditure for refrigerators and freezers		84.94(407.72)	87.95(415.33)	43.61(280.36)	89.08(418.88)	43.83(270.16)
Total expenditure for microwave ovens		14.10(86.98)	14.49(89.23)	8.75(45.82)	14.57(89.83)	9.38(50.58)
Total expenditure for cooking appliances		81.79(413.95)	78.13(401.14)	132.05(558.72)	80.02(407.87)	99.37(469.86)
Total expenditure for washers and dryers		79.45(364.01)	78.87(362.61.14)	87.40(382.73)	79.73(366.60)	76.66(337.32)
Total expenditure for dishwashers		30.35(166.95)	30.75(169.14)	24.85(133.22)	30.99(170.28)	23.98(129.25)
Total expenditure for room air conditioners, portable humidifier and dehumidifier		22.63(154.45)	23.32(158.11)	13.06(86.69)	23.05(159.28)	18.45(93.81)

*Weighted descriptive statistics

Table 3
Logistic regression predicting the odds of energy poverty and energy poverty adjusting for housing costs

Variable	Categories	Energy poverty			Energy poverty adjusting for housing costs		
		β	SE	Exp(β)	β	SE	Exp(β)
Province of residence	Ontario (ref)						
	Alberta	−1.2047	0.2730	0.2998***	−0.9352	0.2235	0.3925***
	British Columbia	−0.6972	0.2216	0.4980**	−0.5228	0.1829	0.5928**
	Manitoba	−0.8596	0.2161	0.4233***	−0.8644	0.1862	0.4213***
	New Brunswick	0.3909	0.1690	1.4783*	0.3146	0.1508	1.3698*
	Newfoundland and Labrador	0.6910	0.1636	1.9957***	0.5044	0.1500	1.6560***
	Nova Scotia	0.7356	0.1582	2.0867***	0.6528	0.1393	1.9208***
	Prince Edward Island	1.0645	0.1996	2.8993***	0.9081	0.1755	2.4797***
	Quebec	−1.0010	0.2063	0.3675***	−0.5680	0.1632	0.5666***
Lives in a rural area	Saskatchewan	−0.3257	0.1906	0.7220	−0.2579	0.1652	0.7727
	No (ref)						
	Yes	0.7781	0.1612	2.1773***	0.5831	0.1457	1.7916***
Tenure type	Own with mortgage (ref)						
	Own without mortgage	−0.0809	0.1818	0.9223	−0.6142	0.1534	0.5411***
	Rented	−0.3561	0.2792	0.7004	0.0139	0.2225	1.0140
Sex	Female (ref)						
	male	−0.2473	0.1401	0.7809	−0.2362	0.1231	0.7896
Age		−0.0369	0.0073	0.9638***	−0.0242	0.0054	0.9761
Marital status	Married (ref)						
	Living common-law	−0.0922	0.3036	0.9120	−0.3905	0.2387	0.6767
	Divorced	0.3528	0.2885	1.4231	0.4281	0.2288	1.5343
	Separated	0.7180	0.2977	2.0504*	0.5775	0.2403	1.7816*
	Single, never married	−0.0057	0.2983	0.9943	0.1257	0.2213	1.1339
	Widowed	0.2295	0.3010	1.2579	0.2261	0.2425	1.2537
Highest level of educational attainment	High school or below (ref)						
	Some post-secondary	−0.1404	0.1541	0.8690	−0.3050	0.1253	0.7372*
	Bachelor's degree or above	−0.1625	0.2037	0.8500	−0.4376	0.1713	0.6456*
Household type	One-person household (ref)						
	Parents with children	−0.3998	0.3385	0.6705	−0.2016	0.2767	0.8174
	Other	−0.9602	0.2876	0.3828***	−0.4626	0.2397	0.6296
Size of household		0.0002	0.1519	1.0002	−0.0416	0.1337	0.9593
Number of children		−0.2357	0.1725	0.7900	−0.0587	0.1494	0.9430
Internet access	No (ref)						
	Yes	−0.2397	0.1695	0.7869	−0.3170	0.1540	0.7283*
Health care costs (i.e., direct costs to household and private health insurance plan premiums)		0.0001	0.0001	1.0001	0.0001	0.0001	1.0001
		0.0002	0.0001	1.0002	0.0002	0.0001	1.0002
Major health care equipment costs (i.e., hearing aids, wheelchairs and scooters, hospital beds, patient lifts, and CPAP machines)							
Major source of income	Employment (ref)						
	Government transfer payments	0.9071	0.2693	2.4770***	0.6934	0.2072	2.0004***
	Other	0.6873	0.3419	1.9883*	1.0672	0.3264	2.9072**
Employment insurance	No (ref)						
	Yes	−0.6277	0.2295	0.5338**	−0.3079	0.1902	0.7350
Full-time earners		−0.8073	0.2186	0.4461***	−0.5223	0.1935	0.5931**
Part-time earners		−0.2543	0.1697	0.7755	−0.0649	0.1335	0.9371
Full-time work weeks		−0.0140	0.0057	0.9861***	−0.0080	0.0046	0.9920

(Continued)

Table 3
(Continued)

Variable	Categories	Energy poverty			Energy poverty adjusting for housing costs		
		β	SE	Exp(β)	β	SE	Exp(β)
Part-time work weeks		0.0095	0.0052	1.0095	0.0032	0.0046	1.0032
Private use vehicle	No (ref)						
	Yes	−0.1720	0.2002	0.8420	−0.4791	0.1632	0.6193**
Private use vehicle cost		0.0001	0.0001	1.0001	0.0001	0.0001	1.0001*
Public transit cost		−0.0002	0.0002	0.9998	−0.0003	0.0002	0.9997
Dwelling type	Single detached (ref)						
	Double (semi-detached); row or terrace (townhouse); duplex	−0.5777	0.2272	0.1860*	−0.6632	0.1938	0.1423***
	Low-rise and high-rise apartments	−1.6819	0.2708	0.5612***	−1.9497	0.2304	0.5152***
Construction period	1961–1980 (ref)						
	1960 or before	0.3303	0.1569	1.3913*	0.2515	0.1346	1.2859
	1981–2010	−0.0011	0.1730	0.9989	−0.0303	0.1376	0.9702
	2011 or after	−0.3697	0.3560	0.6909	−0.3673	0.3361	0.6926
Dwelling needs repair	No (ref)						
	Minor repair	0.2145	0.1502	1.2392	0.3849	0.1277	1.4695*
	Major repair	0.2523	0.2174	1.2869	0.4388	0.1742	1.5509***
Repairs and maintenance—Interior walls and ceilings		−0.0003	0.0003	0.9997	−0.0003	0.0003	0.9997
Repairs and maintenance—Caulking and weather stripping		−0.0015	0.0014	0.9985	−0.0001	0.0005	0.9999
Repairs and maintenance—Insulation		0.0021	0.0009	1.0021	0.0014	0.0008	1.0014
Repairs and maintenance—Roofs		−0.0001	0.0002	0.9999	−0.0001	0.0001	0.9999
Repairs and maintenance—Exterior walls, siding, soffits, fascia, chimneys and foundations		0.0001	0.0001	1.0001	0.0001	0.0001	1.0001
Repairs and maintenance—Windows and exterior doors		0.0001	0.0003	1.0001	0.0001	0.0003	1.0001
Repairs and maintenance—Heating and/or air conditioning equipment		−0.0001	0.0002	0.9999	−0.0002	0.0003	0.9998
Tenants' repairs and improvements		0.0003	0.0001	1.0003*	0.0001	0.0001	1.0001
Total expenditure for refrigerators and freezers		−0.0008	0.0003	0.9992	−0.0007	0.0002	0.9993
Total expenditure for microwave ovens		−0.0010	0.0008	0.9990	−0.0006	0.0006	0.9994
Total expenditure for cooking appliances		0.0006	0.0002	1.0006***	0.0004	0.0001	1.0004**
Total expenditure for washers and dryers		0.0003	0.0002	1.0003	0.0001	0.0001	1.0001
Total expenditure for dishwashers		−0.0001	0.0005	0.9999	0.0001	0.0004	1.0001
Total expenditure for room air conditioners, portable humidifier and dehumidifier		−0.0007	0.0009	0.9993	0.0001	0.0003	1.0001
Constant		1.2469	0.6340	3.4795***	1.1432	0.5064	3.1368*
Nagelkerke R ²		0.171			0.113		
Model N		10,216			10,216		

*p < 0.05, ** < 0.01, *** < 0.001.
SOURCE: *Survey of Household Spending* (2016).

energy poverty and energy poverty adjusting for housing costs, respectively. The first set of explanatory variables relate to employment, income, and expenditure-related variables. Full-time earners are 55% less likely to experience energy poverty than others. Also, households receiving government transfers are 2.48 times more likely to experience energy poverty than those for whom employment is the major source of income. Respondents receiving EI, compared to those not receiving, are also 47% less likely to experience energy poverty. When adjusting for housing costs, having a vehicle for private use is associated with a 39% decrease of being in energy poverty. Similarly having access to the internet is associated with a 28% decrease of experiencing energy poverty.

Next we look at geography and tenure type. All Atlantic provinces exhibit a positive relationship with energy poverty compared to the reference category of Ontario; the odds of being in energy poverty in all other provinces are less likely than in Ontario. Households located in rural versus urban areas are 2.17 times more likely to experience energy poverty. Households owning without a mortgage are 46% less likely to be in energy poverty than those owning with a mortgage (reference category), adjusting for housing costs. Accordingly, households in semi-detached dwellings, townhouses, duplexes, or apartments, as well as those living in low- and high-rise apartments, are less likely to experience energy poverty than households living in single-detached dwellings.

Respondents living in older buildings are at greater risk of energy poverty. Specifically, energy poverty increases by 139% for households living in dwellings built in 1960 or before, compared to those living in dwellings built after 1960. When adjusting for housing costs, the odds of energy poverty substantially increase when considering minor or major repairs. Dwellings needing minor or major repair are associated with a 147% and 155% increase in the odds of being in energy poverty compared to dwellings not needing repair.

With respect to demographics, when adjusting for housing costs, having some post-secondary education or a bachelor's degree or above is associated with 27% and 36% lower odds of being in energy poverty compared to having high school or less. Previous literature indicates the likelihood of experiencing energy poverty is enhanced for women and other marginalized groups (e.g., Robinson 2019; Martiskainen et al. 2021). However,

we did not find significant sex differences. With respect to marital status, those who are separated are 2.05 times more likely to experience energy poverty than those who are married (reference group); no other marital status groups were found to have statistically significant relationships.

Given independent variables are measured in different units, standardized coefficients are useful for comparing the relative strength of different independent variables in a regression. In our model, major income source has the strongest relationship with risk of energy poverty, followed by the number of full-time earners and, lastly, living in rural areas.

Finally, as shown in Figure 3, our Engel curve analysis demonstrates that the relationship between the budget share spent on energy expenditures and household income differs according to whether a household is or is not in energy poverty. For households in energy poverty, at lower levels of household income, income increases are associated with increases in the budget share allocated to energy expenditures, meaning with more overall income, more of it was spent on energy services. For this group, the data are clustered around an income level of \$25,000. Also with this group, the budget share appears to stabilize around just under \$50,000, at which point households in energy poverty are allocating approximately 16% of their total expenditures to energy expenditures. For those not in energy poverty, there is a slight rise in slope at the lower levels of income, and a slight decrease in slope at the higher levels of income, but the overall budget share spent on energy expenditures is relatively stable across all levels of income. The data are clustered around an income level of \$70,000. This group, on average, appears to be spending just over 3% of its total expenditures on energy expenditures.

Concluding discussion

We estimated the share of Canadian households experiencing energy poverty by analyzing data from Statistics Canada's nationally representative SHS. Our findings, based on the widely used (yet also contested) indicator of a household being in energy poverty if they use more than 10% of their income on energy costs, show that between 7% and 9% of Canadians live in energy poverty. Our findings are similar to calculations for England and France that ranged from 10% to 13% (DECC 2016; Imbert et al. 2016) when the 10% indicator was used.

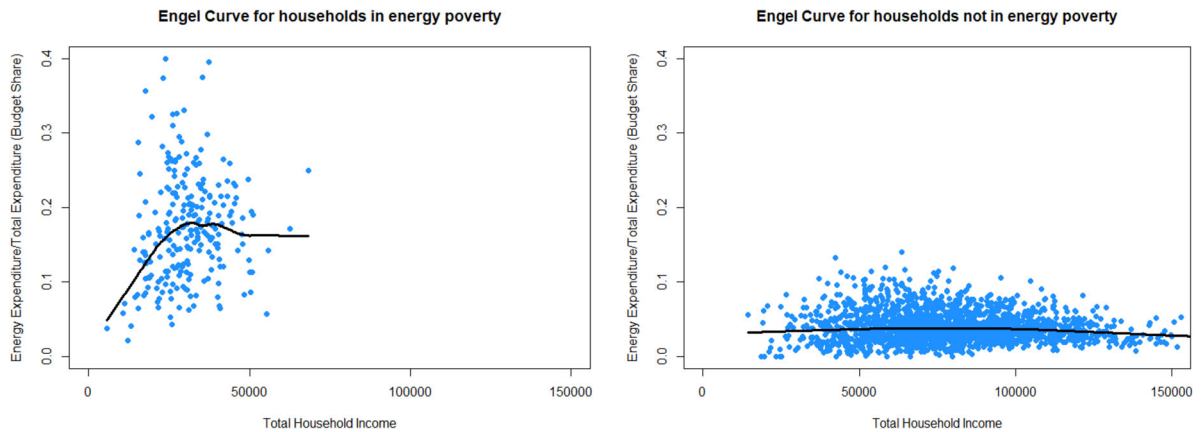


Figure 3

Engel spending curves for energy.

SOURCE: Author's own calculation from the 2016 *Survey of Household Spending*.

Income-related variables are the most significant predictors of energy poverty, with households in energy poverty having lower levels of income compared to those not in energy poverty. Low income has been well documented as an energy poverty predictor (Thomson et al. 2017), as has unemployment (Belaïd 2018). However, having high income does not preclude households from being in energy poverty. Homeowners, for example, usually bear higher home maintenance costs and can experience energy poverty (Devalière 2009; Legendre and Ricci 2015).

We also find that geography matters. First, those living in the Atlantic provinces, where spending on energy is higher, are substantially at higher risk of being in energy poverty than households in other provinces. Second, households in rural areas are more likely to face energy poverty than urban households, as found in other research (e.g., Robinson and Mattioli 2020).

As in other research, dwelling conditions and energy inefficiency—gauged via dwelling age and repair requirements—were associated with higher levels of energy poverty (e.g., Rana et al. 2021). Retrofitting the Canadian building sector is therefore critical. Not only will targeting inefficient buildings reduce energy costs, the co-benefits of health and comfort, which are lacking for some people in energy poverty, would be beneficial (Hoicka and Das 2021).

Additionally, we note important findings by examining the relationship between a household's budget share on energy expenditures and their

household income. First, households that are in energy poverty spent a greater share of their budget on energy services (almost five times more) than households not in energy poverty. Second, for those in energy poverty, spending on energy as a proportion of total spending increases proportionally with increases in household income before stabilizing. In contrast, for those not in energy poverty, the budget share on energy is relatively stable across all levels of income. For households that are in energy poverty, increased energy spending suggests households may be doing without adequate levels of energy services.

Although far from perfect, the 10% indicator has the potential to communicate the important challenge of Canadians' energy costs-to-income ratio, especially in the context of major events such as a pandemic and increasing climate-induced challenges such as heat waves. These will result in differing energy behaviours and needs. For example, more people may work from home and need to use more energy at home. Also people may need to use more cooling devices and systems than previously accounted for in their household budget. As a result of COVID-19, the federal government introduced several measures of support (Canada 2021). Similarly many utilities across the country are offering relief by suspending disconnections, offering payment plans, and other assistance (CUSP 2021). Problematically, however, most of these measures have end dates.

A related issue is price variability. Many households may not find energy affordability challenging

most months, but abrupt price increases could lead to hardship. For example, the 2021 BC flooding contributing to variable vehicle fuel prices demonstrates this possibility. Policies like price caps, energy rebates, or fixed price annual contracts could safeguard against price variability. Future research should also consider the study of “transport poverty” or the inclusion of vehicle fuel prices in energy poverty study.

The federal carbon price may also impose additional burdens, particularly with prices projected to increase from \$65/tonne in 2023 to \$170/tonne in 2030 (Gignac and Sawyer 2021). Such price increases have the potential to impact the demand for energy; they also have implications for the welfare of households. Higher-income households may be able to offset the effect of such energy prices, however those with lower or “stagnant” incomes could be adversely impacted (Meier et al. 2013). Revenue recycling approaches, such as direct rebates or income tax reductions, may offer some assistance, but it is still unclear what levels of support these will actually provide.

With more disruptive events that have the potential to affect energy costs borne by households expected to occur, given stressors like climate change, long-term policies ought to be considered to ensure that the welfare of Canadians is not compromised. More broadly, in the regular day-to-day, energy services should be seen as necessary for human needs and capabilities and decision makers ought to consider the energy burden of Canadians.

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